

## GATE CSE Preparation Roadmap

### 1. Programming + Data Structures (DSA)

- C/Python basics
- Arrays, Linked List, Stack, Queue, Trees, Graphs
- Time and space complexity

### 2. Discrete Mathematics

- Logic, Sets, Relations
- Graphs, Counting, Combinatorics

### 3. Algorithms

- Sorting, Searching
- Greedy, Dynamic Programming
- Graph algorithms

### 4. Computer Organization & Architecture (COA)

### 5. Operating Systems (OS)

### 6. Database Management System (DBMS)

### 7. Computer Networks (CN)

### 8. Theory of Computation (TOC)

### 9. Compiler Design

### Parallel Preparation:

- General Aptitude: 15-20 minutes daily

- Engineering Mathematics: Practice alongside subjects

For 1st year CSE:

C/Python -> DSA -> Discrete Mathematics